

Case study

Potential for peat-to-power usage in Rwanda and associated implications



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ARTICLE INFO

Article history:

Received 11 March 2015

Received in revised form

3 December 2015

Accepted 6 April 2016

Available online 26 May 2016

Keywords:

MESSAGE model

MAED model

Rwandan energy system

Peat-to-power

Levelized cost of energy

ABSTRACT

The persistent low-level access to modern energy in Rwanda is attributed to lack of investment in the energy sector. Accordingly, the investment in energy resources use is analyzed for electricity supply security. During a study period that spanned the years of 2013–2045, scenarios involving population and average GDP growth rates are used for the analysis of future electricity demand and supply. The introduction to the national grid of electricity from peat-fired plants helps phase out oil thermal power, thus the removal of electricity utility price subsidies. The supply of electricity security margin is also observed to be between 32% and 51%. In general, peat-to-power technology is cost competitive compared to the electricity utility price in Rwanda and the region.

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1. Introduction

Due to its role in the development of the national economy, energy is a key strategic sector in Rwanda. The provision of an adequate energy infrastructure is essential for households, industries, and the delivery of high quality services from health facilities, schools, and administrative offices. Rwanda's vision² emphasizes the need for economic growth through private investments and economic transformation; all of which requires a reliable and affordable energy supply [38]. The energy sector in Rwanda is dominated by electric power, petroleum, and biomass sub-sectors. Non-commercial traditional biomass in the form of wood, charcoal, and residues accounts for up to 57%, 23%, and 6% of the country's total energy balance, respectively. In contrast, hydro-electricity accounts for up to 3% and petroleum products account for up to 11% of all utilized energy supplies [16].

The final energy demand can also be viewed in terms of

household, transportation, industry, and public sector energy consumption. Information on the sectoral final energy consumption pattern in Rwanda is not available from previous studies. However, the only information available in Rwanda Land Use and Development Master Plan report regarding the share of sectoral total energy consumption allocates 90% to households, 5% to transportation, 3% to industry, and 2% to public services [44]. The per capita electricity consumption per year is also very low in the region at 30 kWh compared to neighboring countries including Uganda (66 kWh), Tanzania (85 kWh) and Kenya (140 kWh) [78]. The access to electricity is still a problem to about 15% of enterprises in Rwanda [60]. The transportation sector is the primary consumer of imported petroleum products, especially gasoline and diesel. About 75% of all imported petroleum products is consumed within the transport sector, and the remaining 25% is used for electricity generation [56].

Although the national installed energy capacity has increased dramatically over the last two decades, from 25 MW in 1994 to 112 MW in 2012, it has mainly been comprised of 46% oil thermal power, and 53% from hydropower. There are also 2% from natural gas (methane) thermal power, and 0.2% of solar power [17]. Considering the average annual hydropower generation in the region, Rwanda combined with Burundi and the Eastern region of the

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² Rwanda vision (2000–2020) was launched in 2000.

Democratic Republic of the Congo (DRC), they produce only 3% of the total hydropower in the region [13]. In contrast, Ethiopia, Sudan, Tanzania, Uganda, Egypt, and Kenya produce 46%, 16%, 12%, 12%, 8%, and 4%, respectively [13]. The high cost of fuel consumption for thermal power requires the government subsidies to keep electricity retail tariffs in the range of \$0.12–\$0.18 USD, comparatively to the region [17].

Although the electricity access rate has more than tripled from 5% in 2005 to 18% in 2014, the average rate of electrification is still very low as compared to values of 40% and 31% for Africa and Sub-Saharan Africa, respectively [17]. The electricity is transmitted and distributed using high-voltage cables of 110-kV and 70-kV capacities facilitate the total electricity network length of 370 km, where 253 km is of 110-kV capacity and 96 km is of 70-kV capacity [15]. Three major challenges can be identified in the Rwandan electricity sub-sector; (1) electricity demands being nearly equal to the available capacity (which hardly allows for any security reserve margin), (2) costly expenses associated with oil thermal plants, and (3) insufficient investment finances [16].

The objective of this study is the formulation of a medium- to long-term Rwandan electricity system corresponding to demand forecast, utilizing all potential resources available in the country including peat-to-power technology being optimally utilized as a complement to natural gas and renewable energy resources.

2. A review of previous study

Few studies have explored the energy potential in Rwanda. Safari [7] presented a review of existing energy resources and applications in Rwanda, while Gupta and Sood [19] analyzed energy sources in Rwanda and proposed alternative solutions that addressed the lack of fossil fuels. The potential of wind energy is still under investigation; therefore, Safari and Gasore [6] recommend that prior to any decision being made on wind harvesting in Rwanda, further investigation into the details and systematic analysis of speed patterns should be performed. Pigaht and Robert [35] studied the development of micro-hydropower projects as a source of on-grid and off-grid electricity via private sector participation in Rwanda. In general, the potential electricity generated from peat-to-power technology as a facet of energy security has been poorly addressed by previous literature; as it accounts less than 0.1% of world energy supply [52].

3. Peat resources in Rwanda

Rwanda is reported to have peat resources that could be converted into energy. A master plan incorporating peat-derived energy, prepared by Ekono [14] indicates potential peat reserves that amount to 155 Mtons of dry peat, distributed over an area of about 50,000 ha and equivalent to 500 MW electrical powers during 30 years. Vitikka and Lahtinen [1] report the same amount of quantity after performing the peat master plan update study (see Fig. 1 and Table 1).

Peat is generally considered to be the first developmental stage in the coalification process, in which there is a gradual increase in the carbon content of fossil organic material and a concomitant reduction in oxygen [31]. It is also treated as a sub-category within brown coal and defined as a combustible soft, porous or compressed fossil sedimentary deposit of vegetal origin; which is easily cut, light to dark brown in color, and holds a high water content (up to 90% in the raw state) [26]. It therefore shares many similarities with coal, it is used to generate electricity, and it has applications in industrial, residential and other sectors like coal [63].

Peat has played a major role in Finland, Sweden, Indonesia, Ireland, Russia, and various areas of the former Soviet Union [64]. In

Ireland for example, peat has been used for electricity generation since 1950s, and it is estimated at 10% of the total installed capacity [30]. The large-scale use of peat as an energy source has yet to be established in Rwanda. The Peat Energy Company (PEC) is an industrial scale company operating in the southwestern region of Rwanda, and it produces 13,000 ton per year with wet peat to be supplied to the Rwanda Cement Company (CIMERWA) in the region [4]. The PEC initiated mechanical peat production in 2008; however, it is still vulnerable to its limited capacity to sustain large-scale peat production that could handle up to 50 MW of power [50]. Therefore, more investors are also needed in peat mining for use in electricity generation.

4. Methodology and data input

4.1. Energy modeling tools

The Model for Energy Supply System Alternatives and their General Environmental Impacts (MESSAGE) is employed in this study as an optimization model of electricity generation in Rwanda. MESSAGE is designed to formulate and evaluate alternative energy supply strategies consonant with user-defined constraints (such as limits on new investment, fuel availability and trade, environmental regulations, and market penetration rates for new technologies), with objective function to minimize the total energy system cost, as subject to a set of pre-specified constraints [23]. It also aims to determine the feasible least-cost solution for energy supplies or technologies that satisfy future energy demands.³ Electricity is non-storable energy form; therefore, the electricity generation and consumption are considered together because the generation has to follow the required variable demand hour by hour. In this analysis, we perform the modeling of the hourly load curve of Rwanda future electricity demand using the Model for Analysis of Energy Demand (MAED). The data resulted from MAED serves as final demand input to MESSAGE.

Chateau and Lapillonne [3] originally developed the general approach to the MAED methodology and the International Atomic Energy Agency (IAEA) improved it in an attempt to establish required parameters to be specified as input data and equations used for calculating the energy demand of certain sectors [24] [22]. MAED has advantages of evaluating future energy demand based on medium- to long-term scenarios of socio-economic, technological and demographic development. It is also built with detailed considerations of technologies that allow for modeling of the impact of distinct and well-defined technologies on the long-term development of energy consumption [46]. MAED relates systematically to the specific energy demand for producing various goods and services identified in the model, to the corresponding social, economic and technological factors that affect this demand.⁴ Energy demand is disaggregated into a large number of end-use categories; each one corresponding to a given service or to the production of a certain good.

4.1.1. Descriptions of scenarios

The year 2012 represents the base year, whereas the year 2013 represents the starting year in the optimization model process. MESSAGE recommends a base year followed by the model years each separated by one space and the interval between the model years can vary but it should be constant or increasing [23]. Accordingly, the study period of future electricity supply

³ See Ref. [30] for more detailed description of the models internal workings.

⁴ See MAED details at http://www-pub.iaea.org/MTCD/publications/PDF/CMS-18_web.pdf.

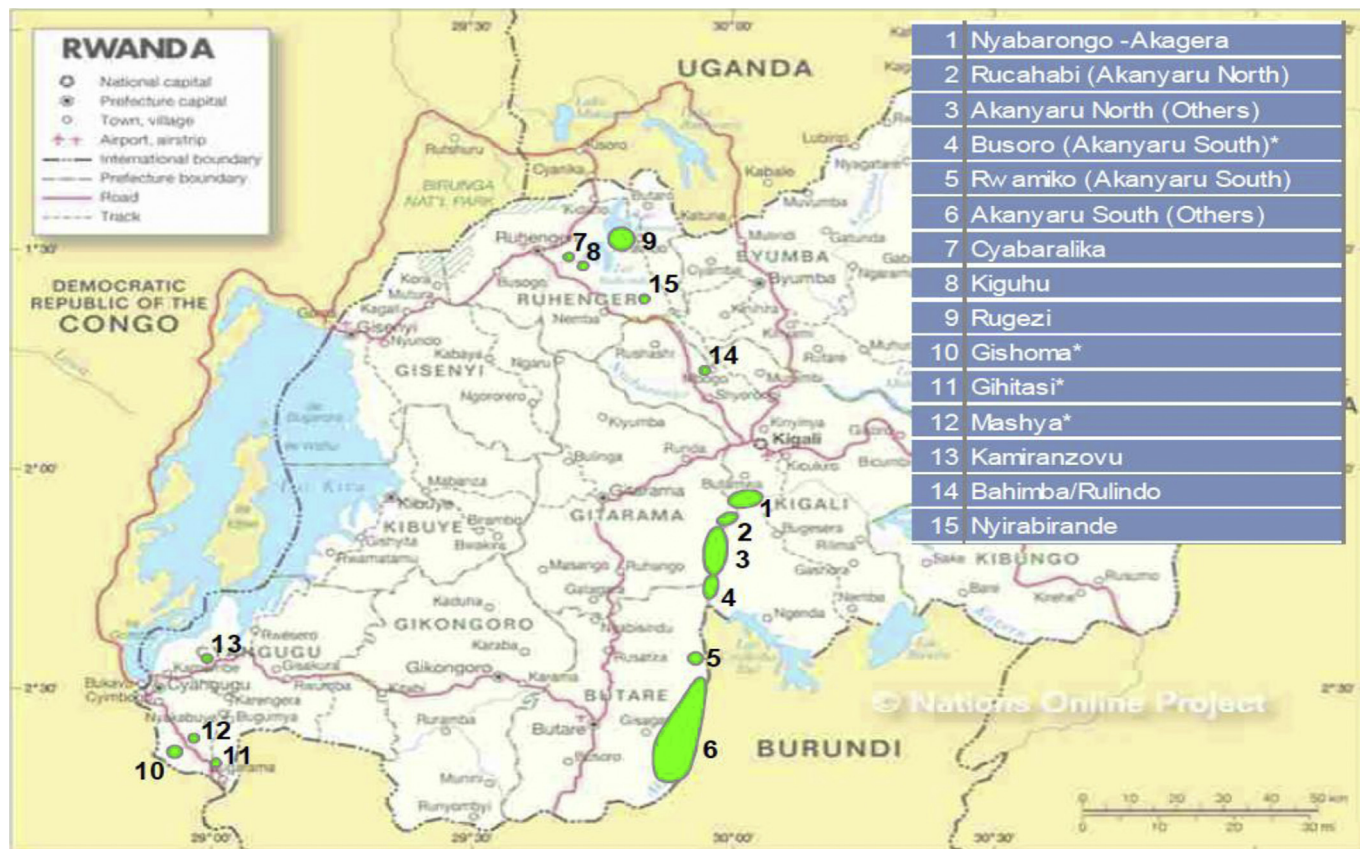


Fig. 1. National peat map [1].

Table 1

Peat reserves estimated in 1992 as updated in 2012 [1].

Location	Scientifically defined peat reserves survey				
	Area (ha)	Thickness (m)	Ash content (%)	Reserves (Mm3)	Reserves Dry matter (1000 t)
Nyabarongo–Akagera	26,740	2–4	9–20	800	40,000
Rucahabi (Akanyaru North)	460	2–4/6	7–13.6	20	994
Akanyaru North (Others)	5120	2–6	10–20	200	10,000
Busoro (Akanyaru South)	800	1–5	6–15	32	1000
Rwamiko (Akanyaru South)	130	1≥20	6–8	14	675
Akanyaru South (Others)	7070	2≥20	6–15/20	920	69,000
Cyabaralika	23	1–5	5.3–17.8	0.4	150
Kiguhu	49	1–4	8.5–14.4	2	240
Rugezi	6500	1≥11	2–15	650	32,000
Gishoma	410	1–5	6.3–13.6	8	463
Gihitasi	80	1–2	14.4	0.6	40
Mashya	30	1–5	2.7	0.8	60
Kamiranzovu	820	1	6.0	8	500
Total	48,200			2700	155,000

encompasses the period of 2013–2045; with a time step of 1 year for the first model period of 2013–2014, 3 years for the period of 2014–2017 and 2020, and 5 years for the period of 2020–2045. With aim to model final electricity demand and electricity supply, the scenarios and assumptions are developed to realize an optimal supply with respect to the most cost effective national supply structure over the study period.

4.1.2. Demand scenarios

We use MAED to perform the electricity demand forecast by choosing 2010–2045 as our study period. The year 2010 is the base

year to present the economic and energy background of Rwanda. The reason for the choice is that 2010 is also the base year of the wide country's energy policy and strategy and it is also during that time the first Power System Mast Plan was published. Furthermore, it is a half time of the country's 2020-vision implementation plan [38]. The gross domestic product (GDP) and GDP components share information (see Table 2) is used to generate data into MAED.

We assume the electricity gross consumption at 279 GWh, and access to electricity for households is 6%, whereas energy consumed by households is 38% [41]. The population of Rwanda is assumed to increase from 10.7 million and reach 13.3 million in

Table 2
Share of GDP by sector [47].

	Agriculture	Industry	Services
2006	38%	14%	42%
2007	34%	14%	46%
2008	33%	15%	46%
2009	34%	15%	46%
2010	31%	16%	47%

2022 and 16.5 million in 2032, before doubling in 2047 and the urbanization rate is assumed to shift from 16.5% in 2012 to 30% in 2032 [37]. The Rwandan GDP has rebounded with an average growth rate of 7%–8% since 2003 [10]. We therefore use the above information to perform the modeling of future electricity demand basing on two scenarios. One is business as usual (BAU) scenario, which says that 8% of the economic growth trends of the past and the development of other factors influencing energy consumption increase at the same trends as the past. The second scenario, which is high electricity consumption (HEC) scenario, we assume that the economic growth rate is increase to 9.5% and the population growth rate also decreases from 2.4% to 2.1%; therefore, the development rate of other sectors influencing electricity consumption also increases.

4.1.3. Supply scenarios

In Rwanda, the electricity was only supplied by hydropower from the 1950s until 2004. The oil thermal power has been connected to the national grid since 2005 as a strategy to solve the problem of electricity shortage in the country. Its supply capacity was increased up to 46% of the total national electricity supply. The hydropower supply decreased continuously between 2000 and 2007 (see Fig. 2). This decrease is attributed to a decline in generation capacity at Ntaruka plant, which used to supply 90% of the total country's electricity consumption. The reduced capacity of Ntaruka hydropower plant was caused by the factors including reduced precipitation, poor management of the main upstream Rugezi wetlands, the headwaters of the watershed, and degradation of surrounding Rugezi-Burera-Ruhondo watershed due to human activity [21].

Although associated with a hydro resources potential to supply

313 MW, currently the overall hydropower installed capacity is only 64.5 MW [40]. Solar energy is another potential renewable resource in Rwanda, however, it has only been supplying 0.4 MW since 2007 [42]. There is no wind energy identified [6]. Furthermore, Rwanda is reported to have potential energy resources such as methane gas, geothermal, and peat, which can also be exploited to increase the current electricity supply. There are 50 billion cubic meters of methane gas equivalent to 40 million tons of oil equivalent (TOE), which could generate an estimated 700 MW of power; and these reserves are equally shared with the Democratic Republic of Congo [16].

The technical and economic feasibility of methane gas exploitation has been demonstrated since 1963 by the small methane extraction pilot plant at Cape Rubona, 8 km to the southwestern of Gisenyi, and the power plant has worked reasonably well for 40 years [65]. KIVUWATT Power 1, a 4.5 MW pilot plant developed by the Rwandan government, has been operating at a capacity of about 1.5 MW since late 2007 [43]. The Rwanda Energy Company, a subsidiary of the Rwanda Investment Group, has developed a 3.6 MW capacity of methane thermal plant, while KIVUWATT power 2, a subsidiary of Contour Global (from the United States of America), is expected to develop a 100 MW plant [41].

Furthermore, the geothermal energy resource potential in Rwanda is estimated to 740 MW from the combined regions of Kalisimbi, Gisenyi, Kinigi, and Bugarama, which possess 320 MW, 200 MW, 120 MW, and 60 MW of energy, respectively. Other regions, aside those listed above, generate 40 MW [1]. The above geothermal sites were identified by BRGM (Bureau de Recherches Géologiques et Minières) in 1980's with resources temperature in excess of 150 °C suitable for geothermal power generation [65] [53]. The total annual electricity supply by technology (2000–2011) is presented (see Fig. 2).

We therefore assume the potential candidates for energy resources of electricity supply as described in Table 3. It is important to mention that the optimal capacity addition required for renewable energy candidates such as hydropower, solar, and geothermal energy is performed. The oil thermal power plants have been installed since 2005 and reported to be costly to operate; consequently, they increased the electricity tariffs by over 100% and reduced the country electricity competitiveness in the region [65]; therefore, they are assumed to be phased out by 2017.

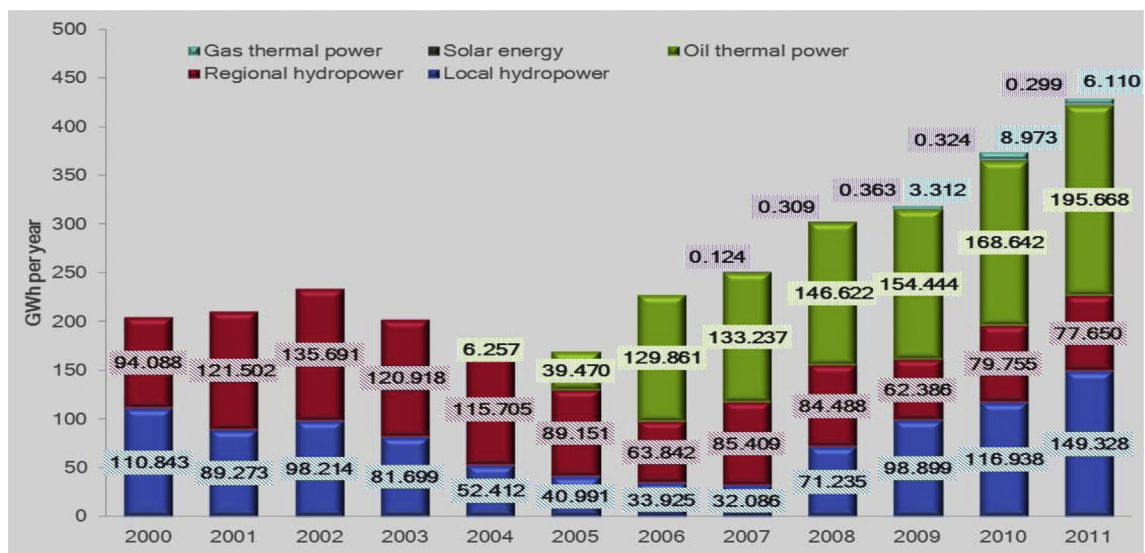


Fig. 2. Total electricity supply by technology [16].

Table 3
Potential resources for energy supply.

Energy resources	Supply scenario in MW
Methane gas thermal power	350
Peat-to-power	300
Local and regional hydropower	350
Micro-hydropower	150
Geothermal thermal power	300
Solar energy	200

4.1.4. Existing and potential candidates Rwanda's electricity system in Rwanda energy system

The Rwandan electricity system is described as a part of national energy system including all energy sources (see Fig. 3). It is structured in a supply network that includes final, secondary and primary, and domestic energy resources (peat, and natural gas) which are termed “levels.” The linking of energy levels is realized using conversion technologies (resource extraction, treatment, import to primary, generation to secondary, transmission, and distribution to final and meeting demand) that consider both currently available and future candidates. The vertical lines represent energy forms at each level (for example, natural gas and peat in resources and primary levels, oil and traditional fuels in primary, electricity in the secondary, and final/demand, etc.).

The blocks represent technologies utilized for the specific energy form to be transformed. The horizontal lines with directions represent the flow of energy input and output forms, carried by relevant technologies between levels. For example, this includes the use of oil energy as an input, the utilization of Oil_PP technology to generate electricity as output in the secondary energy level, and the import of oil resources utilizing Oil_Import technology. Gas_PP technology uses natural gas fuel as an input to generate electricity,

which is an output at the secondary level, while Peat_PP technology uses peat resources to generate electricity. Hyd1 technology represents local hydropower, Hyd2 technology represents regional hydropower, Solar_PP technology represents solar energy, and Geothermal technology represents geothermal energy; all of these types generate electricity at the secondary level.

4.1.5. Economic features

Peat-fired power plants are assumed to be located less than 15 km from the source peat bogs to reduce the cost of raw peat transportation. The steam generator used in these plants is based on a fluidized bed technology (CFBC/AFBC) single drum, natural circulation, water tube, balanced draft, top-supported, and semi-outdoor type design to cater to duty requirements. The capacity of the plant is 15 MW or 50 MW, depending on the amount of peat available in each peat bog location. The thermal efficiency of lignite coal, in which category peat is qualified, is estimated at 41%; however, for Rwanda case, as it is a new technology, we roundly set it at 40% [32]. The overnight cost is estimated to be US\$2010/kW and it corresponds to the overnight investment costs presented for lignite (i.e. lowest rank of coal with same characteristics as peat) power plant in the reviewed studies (i.e. [12] [66–69]).

Depending on the capacities of hydropower plant operation in Rwanda, this energy type is divided into two categories by the present study; Hyd1 and Hyd2, respectively termed local and regional hydropower technologies. Hyd1 and Hyd2 are estimated at 100% of power generation efficiency and the overnight costs are estimated at US\$2990/kW and US\$1910/kW for Hyd1 and Hyd2, respectively [27] [13]. Other renewable energy sources, such as solar and geothermal types, are respectively estimated at US\$3540/kW and US\$2620/kW, and 100% and 15% of plant efficiencies, while, the plant efficiency is 37%, and the overnight cost is estimated to be US\$700/kW for Gas_PP [27] [13]. The operation time of gas power

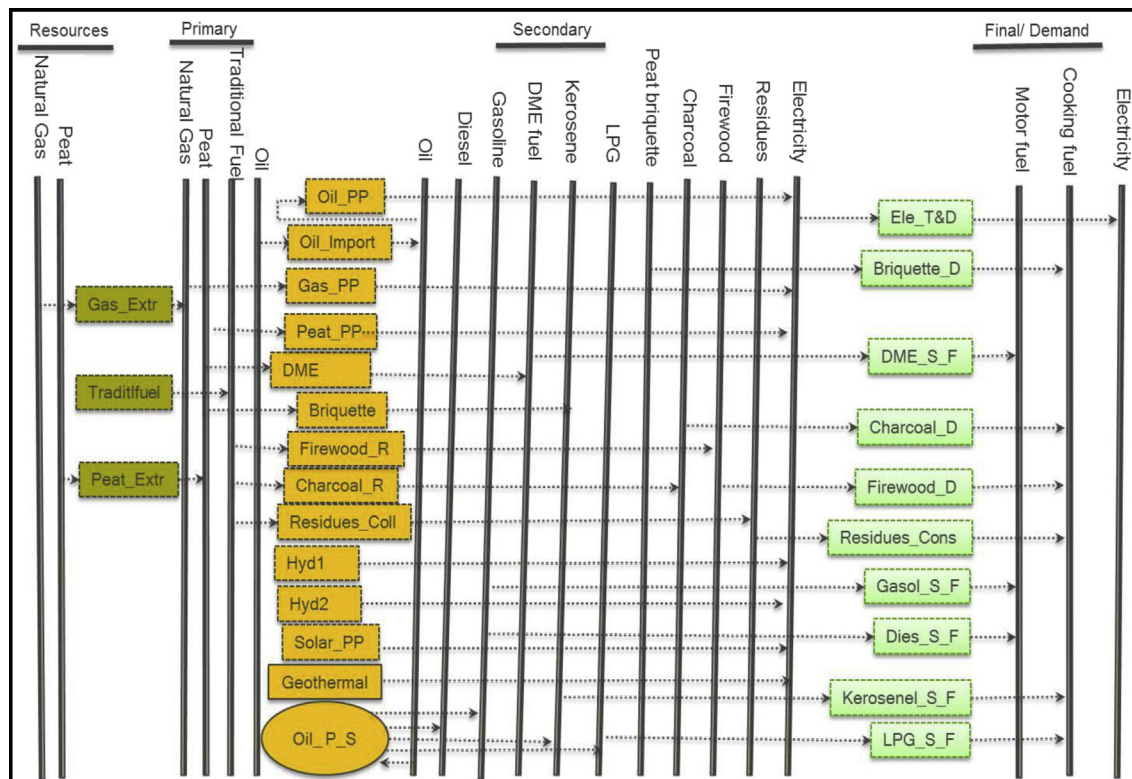


Fig. 3. Flowchart of Rwanda's MESSAGE reference energy system.

plants are assumed to be at 80%, corresponding to a total of 6288 h, whereas, for peat power plants are estimated at 91% (7150 h). All hydropower operation time is set at 50% (4380 h), that of oil power at 80% (6288 h) and geothermal at 70% (6132 h), and that of solar energy is 18% (1577 h) [27].

The overnight cost of transmission and distribution is estimated at US\$1120/kW after compilation and analysis of data on the investment required of electricity supply, available in Rwanda electricity master plan [18] and electricity access program [9]. The transmission and distribution operation time is 98%. The economic lifetime (plant life) of power plants is the period during which the power plant investment should pay itself back, although the technical lifetime is usually longer. The economic lifetime of hydro-power plants is fixed at 50 years, and the transmission economic lifetime is 60 years. Solar, geothermal, oil thermal, gas-fired and peat-fired plants are fixed at 25 year each [27]. Table 4 summarizes the techno-economic features assumptions for our model.

The cost increase of energy technology establishment is primarily caused by increases in construction costs, the price of construction materials and power plant components, as well as the high cost of importation (as necessary in a landlocked country like Rwanda). Furthermore, the cost of Gas_PP and Peat_PP technologies are also increased by the extraction technology, which is imported.

In this study, the price level of September 2014 is used in the calculations. The electricity generation efficiency is expressed as the annual efficiency corresponding to the average over an entire year. Since 2004, the government of Rwanda has spent over FRW 60 billion, estimated at US\$40 million annually, subsidizing electricity consumers to cover the high cost of fuel consumption for oil thermal power [17], however, any possible state subsidies is not taken into account in this formulation. Natural gas power and peat-to-power plants are assumed to be located near existing raw material deposits, such that its transportation does not contribute significantly to the investment cost. The combined cycle gas turbine plant consists of a gas turbine, a waste heat boiler, and a steam turbine. On the reason of assessing the different investment costs of energy technology over the study period, the levelized cost of electricity (LCOE) indicator is utilized. The concept and methodology of LCOE assumes that the life cycle presents worth value of all revenues produced by generated electricity equals the present worth value of all expenditures incurred in plant implementation and operation [25].

Energy policy analysts and project evaluators often use LCOE to develop first-order assessments of a project's attractiveness, and to compare alternate technologies [8]. It is important to mention that LCOE methodologies do not focus on a power plant's different impacts on the electricity system as a whole; however, LCOE is focused on the producers and forget the total the total costs that have to be paid by the consumers or society [59]. LCOE employs techniques, methodologies, and conventions applicable for the

comparative evaluation of future plant (supply-side) and demand response (demand-side) alternatives. These costs comprise that portion of per-unit-of-electricity cost incurred over the lifetime of a plant that, when they are discounted and summed, they yield the total present value cost of the invested capital [25]. It is therefore equal to the constant average price at which electricity must be sold to reach the breakeven over the economic lifetime of the power plant [59]. The LCOE can be directly compared to the price of the local utility charges (i.e., if the energy system generates electricity at a cost less than the utility price, then the project is economically feasible) [5].

4.1.6. GHG emissions from energy conversion technologies

Emissions of carbon dioxide (CO₂) from the combustion of peat are calculated to assess the impact of Peat_PP in relation to the total emissions from electricity generation in the country. CO₂ emissions from peat combustion depend on the carbon content of the peat, its heating value and the amount of peat used by the technology [51]. The average data on carbon content of the peat and the peat heat value are obtained from the analytical comparison between the literature of the physical and chemical properties of peat in Rwanda and our experiment results of peat randomly sampled from BISIKA, BAHIMBA, NDONGOZI and NYIRABIRANDE in Rwanda (see Fig. 1). While the carbon content average is 53.6% (S 5.6, range 27.2, n 109), the average Sulfur concentration of peat is 0.69% (S 0.71, range 4.59, n 228) depending on the region (see Table 5).

Sample of 200 different type of peat is collected and before using thermal methods to determine the peat's chemical characteristics, it was firstly naturally dried. The summary of the experiment results is presented (see Table 6). Percentage values with respect to moisture, volatile matter, ash, and fixed carbon are reported during the proximate analysis in line with the methodology published by Sadek and Debot [58]. The method of calculating the heat values by using data derived from ultimate analysis is also employed. Percentage values of carbon, hydrogen, oxygen, nitrogen, and sulfur are all determined by elementary analysis, as well as moisture and ash values are used to calculate calorific values [58].

We use the relationship that occurs during the combustion to calculate the CO₂ emissions; one atom of carbon combines with two atoms of oxygen from the air to produce one molecule of CO₂. The atomic weight of CO₂ is 44, which corresponds to the atomic weight of C equal to 12 and O equal to 12. The formula used to calculate CO₂ emissions from Peat_PP is as follows [12] [70] [71]:

Table 5
Average values of chemical characteristics of peat [49].

Mire and core	C	H	N	O	C/N
Average	53.6	5	1.7	28.3	33
Standard deviation	5.6	0.5	0.3	3.1	7
Range	27.2	2.3	1.4	18.8	39
No. of samples	109	109	109	109	109

Table 4
Techno-economic features for energy technologies.

	Ovrn cost (US/kW)	Fix-O&M cost (US/kWyr)	Variable-O&M cost (US/MWh)	Plant life (year)	Constr time (year)	Oper time (%)	Input	Output	EF (%)
Oil_PP	0	0	160	25	0	0.80	Oil/HFO	Elect	43
Gas_PP	700	6	30	25	3	0.50	Natural Gas	Elect	37
Peat_PP	2010	23	4.0	25	3	0.91	Peat	Elect	40
Hyd1	2990	15	6	50	4	0.50	—	Elect	100
Hyd2	1910	15	6	50	4	0.50	—	Elect	100
Solar_PP	1290	35	0	25	1.5	0.18	—	Elect	100
Geothermal	2550	0	51	25	4	0.65	—	Elect	15
Ele_TD	1120	145	0	60	4	0.98	—	Elect	80

Ovrn cost: overnight cost; Fix-O&M cost: operation and maintenance fixed cost; Variable-O&M cost operation and maintenance variable cost; Constr Time: construction time; Oper Time: operation time; Elect: electricity; HFO: heavy fuel oil.

Table 6
Results from peat analysis.

Sample	Proximate analysis				Ultimate analysis					HHV (kcal/kg)	LHV (kcal/kg)
	(As-received basis, wt. %)				(Dry ash-free basis, wt. %)						
	M	VM	Ash	FC	C	H	N	O	S		
Bisika	76.69	11.09	12.2	0	37.13	4	1.38	56.82	0.67	3060	2857
Bahimba	79.47	12.26	8.18	0.09	37.45	4.33	1.61	55.59	1.02	3360	2664
Ndongozi	88.82	10.17	0.95	0.06	57.36	5.57	1.77	34.92	0.38	5100	4268
Nyirabirande	83.84	13.07	2.92	0.18	61.57	5.8	0.42	31.89	0.32	4580	3860
Sample	Proximate analysis (Dry basis, wt. %)										
Bisika	—	47.6	52.4	0							
Bahimba	—	59.72	39.8	0.44							
Ndongozi	—	90.97	8.5	0.54							
Nyirabirande	—	80.83	18.1	1.11							

$$\text{KgCO}_2 \text{ per net GJ} = (C/100) \times (3664/HV)$$

where C is carbon content in peat (percentage, dry basis from ultimate analysis) and HV is heating value (MJ/kg, dry basis), 1 MJ/kg = 429.9 Btu/lb.

We assume the complete combustion to occur when 1 kg of carbon combines with 2.667 kg of oxygen to produce 3.667 kg of CO₂. We use the carbon content average of 50.38% and heating value of 4025 kcal/kg of high heating value (HHV) and 3412.25 kcal/kg of low heating value (LHV). As a result, on average Rwanda peat emits 109.6 kgCO₂ per GJ (~0.395 kgCO₂/kWh) on HHV basis and 129.23 kgCO₂ per GJ (~0.465 kgCO₂/kWh) on LHV basis.

The CO₂ emission factors of oil in thermal plant and natural gas power plant are also used to specify environmental loadings for their related technologies. In general, the emissions of each greenhouse gas from stationary sources are calculated by multiplying fuel consumption by the corresponding default emission factors, and most of the fuel consumption are estimated from energy use statistics and measured in terajoules (TJ) [11].

We identify the upper and lower limit of the emissions and above the upper limit; the external cost of the emission is employed [35]. The CO₂ emission cost is estimated by reference to the South African emission tax situation. The South African government has implemented CO₂ emissions trading [2], and the South African carbon tax is adapted to the Rwandan case and included in the LCOE.

5. Results and discussion

5.1. Demand forecast and supply

We present the resulted final electricity demand forecasts distributed by scenarios (see Fig. 4). The final electricity demand grows from 279 GWh in 2010–3361 GWh in 2045 for BAU scenario, and from 279 GWh in 2010–4940 in 2045 for HEC scenario. According to scenarios and assumptions defined in previous section, the MESSAGE model is used to formulate an optimal long-term electricity supply strategy that assures the coverage of any given future demand at minimum costs over the study period of 2013–2045.

The electricity demand forecast results are input data into MESSAGE. The achieved predictive results in terms of energy distribution at primary and secondary levels at minimum cost are assumed to cover the modeled specified demand forecast. Oil thermal power plants that are assumed to phased out by 2017 impacts on the oil import volume, as a result the HFO at primary level stops by that time, whereas peat and natural gas, at primary

level, are expected to increase continuously up to 2045 (see Fig. 5).

The Rwanda government is currently developing 15 MW in Gishoma (South Western of Rwanda) while private developers have also committed the development of 200 MW to be commissioned by 2017 [72]. With this figure, we assume the Peat_PP technology to supply 207 GWh (26% of total electricity supply in 2017). However, this figure does not correspond to the above 215 MW to be installed by 2017. Currently, only private investors are developing 80 MW as the first phase of the project [20]; we therefore assume 30 MW of installed capacity to be available by that time (see Table 7).

The technology of Hyd1, Hyd2 and Gas_PP technologies supply 438 GWh corresponding to 38%, 131.4 GWh (12%) and 31.4 GWh (4%), respectively in 2017 (see Fig. 6). The total installed capacity of Peat_PP technology reaches 60 MW in 2020, with capacity to supply 420.5 GWh (40% of total national electricity supply); while no new investments are envisaged for Hyd1 and Hyd2 technologies. However, the investment of Gas_PP technology increases its electricity supply to 53.4 GWh (5% of total electricity supply).

The Peat_PP technology electricity supply remains at the same level until 2025; new investments increase its capacity to supply 1051.2 GWh in 2030. This continues to increase to 2102.4 GWh in 2045. However, Gas_PP and Peat_PP technologies are almost at the same level of electricity supply, respectively at 31% and 30% in 2040, and 26% and 25% in 2045 (see Fig. 6 and Table 7).

The total supply of electricity at the secondary level increases from 547.8 GWh in 2013 to 8488.6 GWh in 2045 (i.e. to about 14 times of increase). The total installed capacity was 115 MW in 2013, being composed of 47.4 MW of regional hydropower, 15.5 MW of local hydropower, 3.6 MW of natural gas thermal power, 48 MW of oil thermal power, and 0.50 MW of solar energy (see Fig. 6). A considerable amount of MW is installed during the study period, so that the existing and newly installed capacity increases up to 215 MW in 2017 (see Fig. 7). This new installed capacity is different from 563 MW, the currently government target by 2017–2018 [61]. This big difference may be found in new partnerships the country has with regional countries on import of electricity, commissioning of solar photovoltaic technology, which can sometimes take only six months to be installed and commissioned [34].

Although, new investment doubles the capacity of Peat_PP and Gas_PP technologies in 2020, the decommissioning of Oil_PP technology decreases the total installed capacity to 202.6 MW compared to 2017 (see Fig. 7). Due to the peat-to-power technologies, which covers only to around 31% of Oil_PP decommission. Peat_PP is a primary technology in use especially because of its high presentation of operation time than other technologies, while the funding for renewable energy (solar and geothermal) technologies are being mobilized. There is no additional capacity for Hyd1

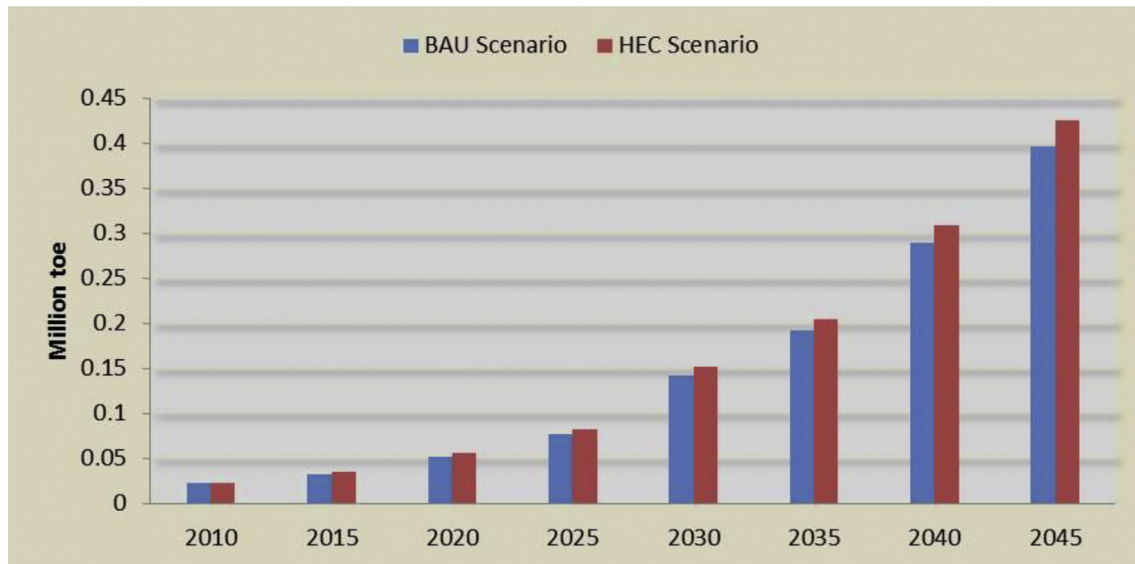


Fig. 4. Forecasted final energy demand.

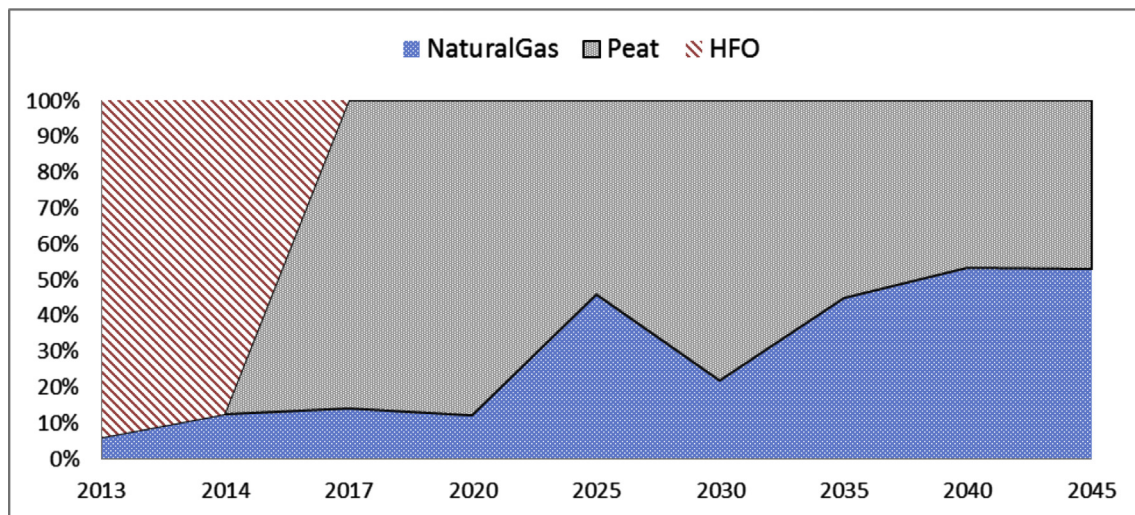


Fig. 5. Distribution of resources by electricity generation.

Table 7

Electricity generation by technology (GWh).

	Oil_PP Elect	Gas_PP Elect	Peat_PP Elect	Hyd1 Elect	Hyd2 Elect	Solar_PP Elect	Geothermal Elect	Total
2013	255.9	15.8	0	207.6	67.9	0.7	0.0	547.8
2014	255.9	15.8	0	350.4	131.4	0.7	0.0	754.1
2017	0	31.4	207	438.0	131.4	0.7	0.0	808.5
2020	0	53.4	420.5	438.0	131.4	0.7	0.0	1044.0
2025	0	328.5	420.5	438.0	400.3	0.7	0.0	1588.0
2030	0	268.9	1051.2	438.0	657.0	0.3	0.0	2415.4
2035	0	1055.7	1401.6	391.9	657.0	0.0	0.0	3506.2
2040	0	1758.2	1660.9	876.0	657.0	0.0	636.2	5588.3
2045	0	2190.0	2102.4	1533.0	657.0	166.5	1839.6	8488.6

Elect: electricity.

technology from 2017 to 2035 (see Fig. 6).

Furthermore, solar energy is likely to be available in 2045 due to its high overnight cost in the optimal solution. However, as solar energy overnight cost is rapidly decreasing with time on the market, its installation can potentially be initiated earlier than the

stated timeframe. This could also be applied to other renewable energy sources depending on the availability of funds for investment. The overall assessment of the country's policy, when it comes to renewable energy plan, we find the target ambitious and impossible to be achieved by 2017 [29]. The example can be found

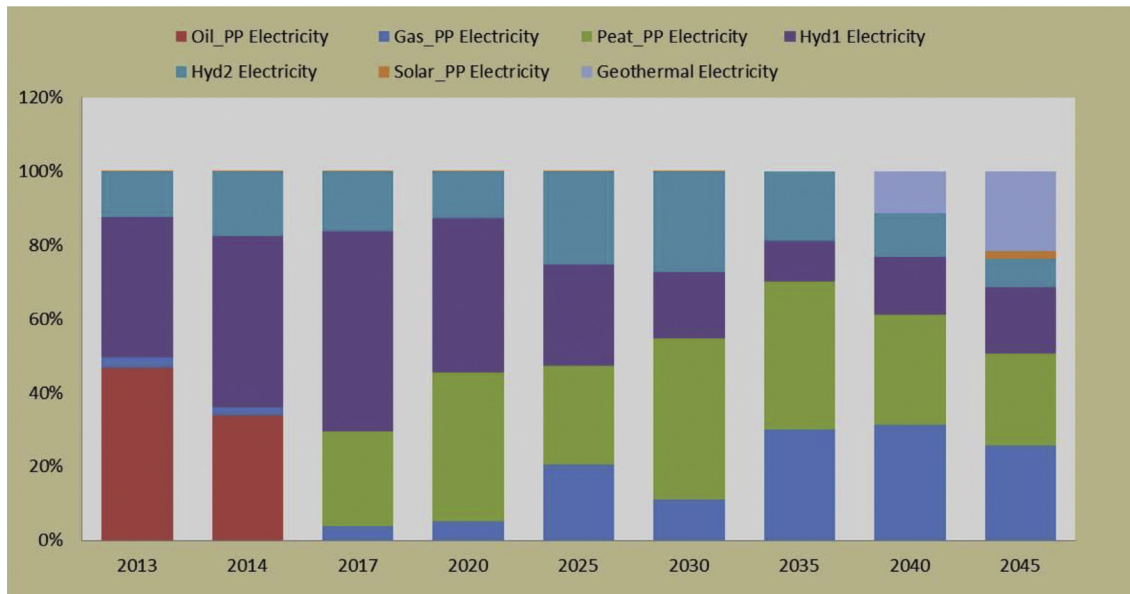


Fig. 6. Share of energy at secondary level.

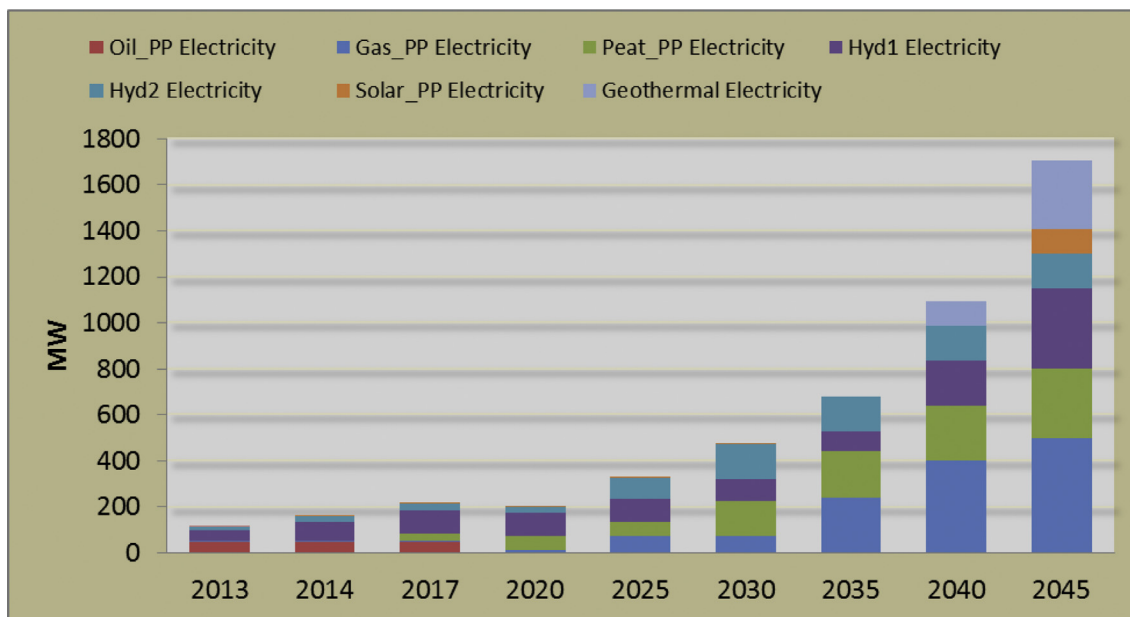


Fig. 7. Total installed capacity by technology.

in the investment of geothermal technology, which is also planned to be available by that time; however, this technology is still under final study to be implemented.

The electricity derived from natural gas and the geothermal potential are assumed with consideration of the total amount of resources available, as specified in the previous section. Considering the available peat resources, Peat_PP technology is seen as a good technology to boost the electricity supply level and especially as an alternative to Oil_PP technology. During the study period, Peat_PP technology remains a reliable technology that helps maintain the balance between electricity demand and supply. Accordingly, the entirety of the newly installed capacity over the study period amounts to 1705 MW, which are capable to supply 8488.6 GWh in 2045.

Generally, by comparison, the results trends of future electricity

supply from our study are not far from electricity generation forecasted by the East African Community [13]. Although, the effort are made by the government to increase the electricity supply in Rwanda, we observe that, in 2030, the total installed capacity of electricity compared to other countries in the East Africa is still at low level like Djibouti and Burundi [13].

5.2. Predicted electricity demand versus potential supply and implication

The security margin of electricity supply is also assessed. The ability of energy technologies to supply electricity to end-users is identified as the difference between demand forecast and potential supply. In addition, the technical losses during transmission and distribution are also deducted from the total supply at secondary

Table 8
Electricity demand vs potential supply.

	Supply at secondary level (GWh)	Predicted demand (GWh)		Security margin before losses (GWh)		Supply security before losses (%)		25% losses (GWh)	Supply at final level after losses 25% of (GWh)	Supply security after losses (%)	
		BAU	HEC	BAU	HEC	BAU	HEC			BAU	HEC
2013	547.8	279	279	269	269	49%	49%	137	411	32%	32%
2014	754.1	326	349	428	405	57%	54%	189	566	42%	38%
2017	808.5	384	411	425	398	53%	49%	202	606	37%	32%
2020	1044	488	522	556	522	53%	50%	261	783	38%	33%
2025	1588	605	647	983	941	62%	59%	397	1191	49%	46%
2030	2415.4	896	958	1520	1457	63%	60%	604	1812	51%	47%
2035	3506.2	1651	1767	1855	1739	53%	50%	877	2630	37%	33%
2040	5588.3	2233	2386	3355	3202	60%	57%	1397	4191	47%	43%
2045	8488.6	3361	4940	5128	3548	60%	42%	2122	6366	47%	22%

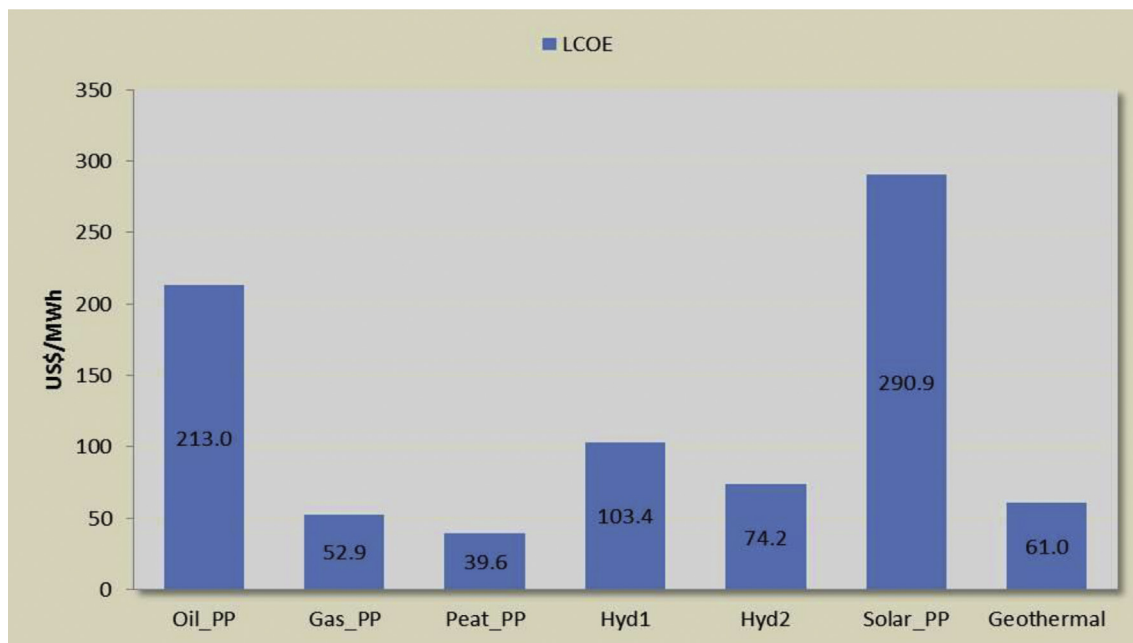


Fig. 8. Levelized cost of electricity at 6% discount rate.

level, and they are estimated at 25% [36]. The security margin is at high level under the BAU scenario in 2014, 2025, 2030, 2040, and 2045 with 42%, 49%, 51%, 47% and 47%, respectively (see Table 8). Under the HEC scenario, the security margin is at high level in 2025 (46%), 2030 (47%), and 2040 (43%). We therefore note that during the study period the both estimated scenarios (BAU and HEC) of forecasted demand is secured at more than 30% supply margin, which looks important for any increase in the electricity demand or unaccounted losses.

The MESSAGE model also reports the results of LCOE (see Fig. 8). The LCOE of Peat_PP and Gas_PP technologies are compared to literature review reported [12] [73–75]. While the LCOE of Gas_PP technology (US\$52/MWh) meets exactly the mean, the Peat_PP (US\$39.6/MWh) looks to be lower than the estimated level. LCOE of Hyd1 and Hyd2 technologies are US\$103/MWh and US\$74.2/MWh, respectively. Therefore, Hyd1 technology looks higher cost than Hyd2. The reason for this feature is found in the capacity to be installed; because most of local hydropower plants are qualified to be small and their overnight costs are higher than regional hydropower and for hydropower plant; more the capacity is less is the overnight cost [13] [28]. The LCOE of geothermal technology is US\$61/MWh, while the LCOE of Solar_PP technology is US\$291/MWh. In general, all renewable energy technology types are more

expensive than Gas_PP and Peat_PP technologies [76] [33].

The instability of macroeconomic factors (i.e. the exchange rate between the Rwandan franc and the US dollar, crude oil prices, and domestic inflation) that drive the energy sector in Rwanda cause frequent increases on country's electricity utility price. This is important to mention because the country's energy system depends largely on importation crude oil used in thermal power plant as well as in transportation sector. LCOE of US\$213/MWh of Oil_PP also confirms the significant impact of the high cost oil and high rental fee paid by the government to oil thermal plant private owners. The literature review also confirms this high cost, although it is lower than what we find here; the surplus identified is caused by including high rental cost in our modeling [13].

In the face of increasing crude oil costs and high rental fee, the Government of Rwanda increased the subsidies of the end-user tariff of electricity up to 40% to keep the electricity tariff in the range of regional tariffs. Therefore, the tariff increased from Rwandan franc (FRW) 112 to FRW134/kWh (about US\$200/MWh)⁵ (value addition tax exclusive) [55]. As a result, aside Solar_PP

⁵ <http://www.xe.com/currencyconverter/convert/?Amount=1&From=USD&To=RWF>.

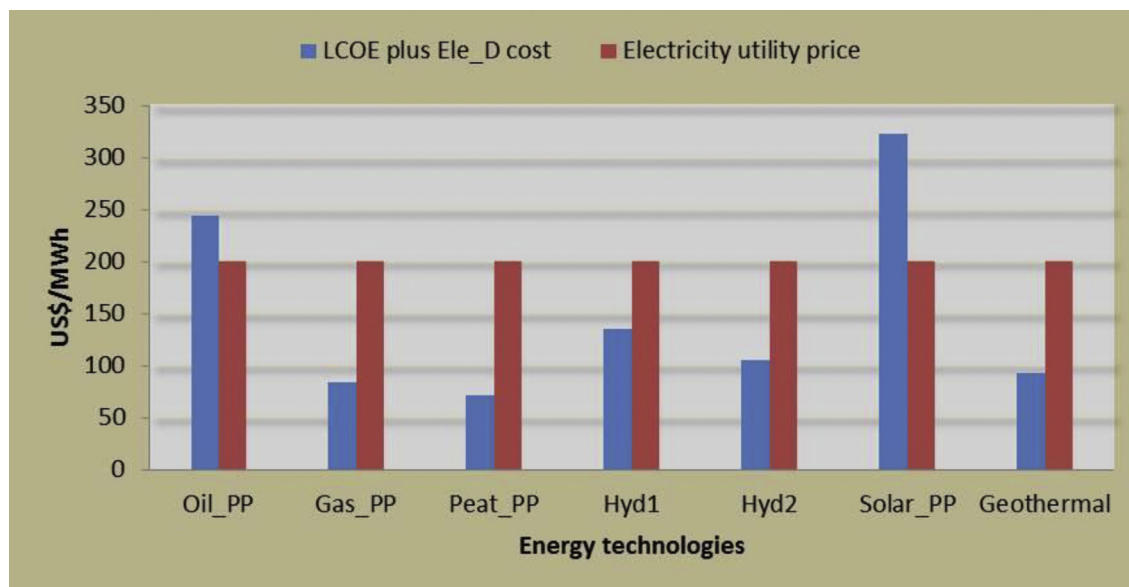


Fig. 9. Comparison of electricity cost level to electricity utility price.

(US\$290.9/MWh) and Oil_PP (US\$213/MWh), other technologies show to generate electricity at a LCOE that is less than the national electricity utility price (see Fig. 9).

For every LCOE, its summation with electricity transmission and distribution costs is also compared to the national electricity utility price to analyze the level of margin. All energy technologies conversion behavior compared to the current electricity utility price remains competitive except Solar_PP with US\$123/MWh more than the utility price. This is caused by high investment and fixed costs, and low capacity factor (23%) [13]. Peat_PP remains the competitive technology for the country, at US\$71.5/MWh with cost margin to the utility price of about 64% (see Fig. 9).

5.3. Forecasted CO₂ emissions from energy technologies

The MESSAGE model calculates the emissions from energy technologies, as based on emission factors calculated in previous section for Peat_PP and average emission factors of Gas_PP and

Oil_PP [11]. We presented the CO₂ emissions trends by technology in relation to total electricity generation (see Fig. 10). With related energy technologies, CO₂ emissions from Peat_PP increase from 96 ktonCO₂ in 2017 (94% of total CO₂ emissions) to 977.6 ktonCO₂ in 2045 (69% of total CO₂ emissions). The total CO₂ emissions during the study period increase from 71.5 ktonCO₂ in 2013 to 1420 ktonCO₂ LHV and 1272 ktonCO₂ HHV in 2045. These CO₂ emissions results compared to 2011, 2012 and 2013 when the Oil_PP and Gas_PP are the main source of CO₂ emissions connected to electricity grid, we find our results to be very low [56]. The reason may be found in different level of electricity generation forecast by oil thermal power plant (391 GWh) [56], against 256 GWh in this study. However, it is in 2020 that the CO₂ emissions reach the estimated 2013 CO₂ emissions. Peat-to-power is identified to be a significant source of CO₂ emissions same as the Gas_PP technology (i.e., the trends increase during the study period as peat power plant planned capacity is fully commissioned). It is important to mention that the above LCOE calculations include the

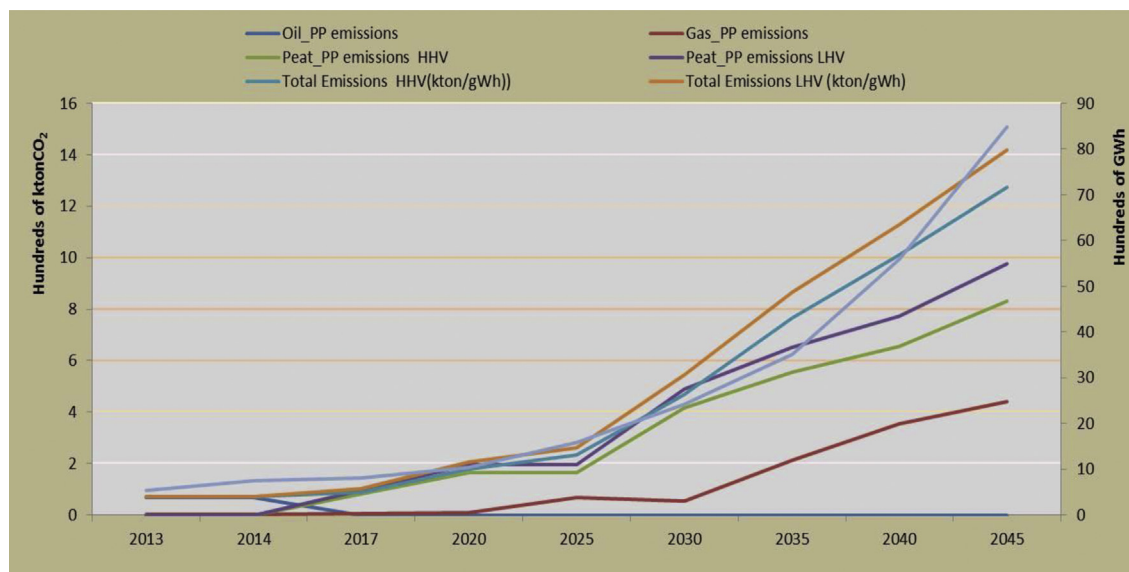


Fig. 10. Comparison of CO₂ emissions by technology to electricity generation.

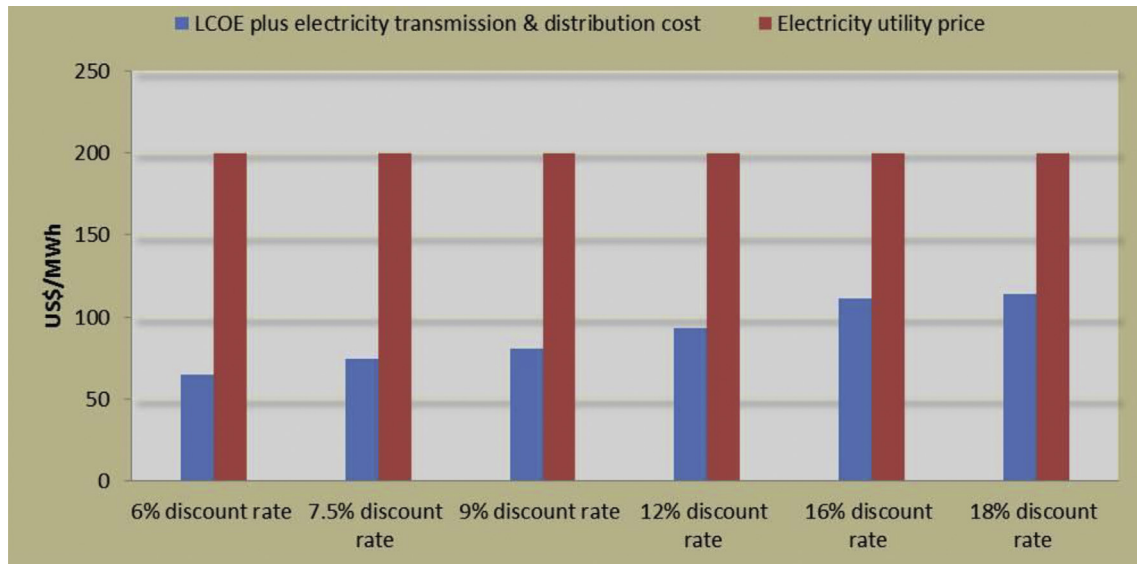


Fig. 11. Comparison of liverized cost of Peat_PP technology to electricity price.

carbon emission price. The average of carbon emission price is estimated to be US\$11.7/tCO₂ [2].

5.4. Sensitivity analysis

We explore the sensitivity of the achieved MESSAGE results of the electricity supply of Peat_PP using three input parameters including discount rate, overnight cost, plant factor, and technology efficiency. The objective of this exploration is to identify the weight of the uncertainty of the above employed parameters and their implications to the investment of Peat_PP technology (see Figs. 11 and 12). In Rwanda, like in any other developing countries, the discount rate is unstable and changes by increasing. The discount rate should be equal to the long-term real interest rate [23], thus, its uncertainty is high. Therefore, we weight the impact caused by its changes on the electricity generation cost of Peat_PP technology. As a result, from the discount rate of 6% as our reference case, we weight the sensitivity results to the alternative values increasing up

to the discount rate of 18%. The results prove that the discount rate is very sensitivity to LCOE, however, this technology remains competitive in the range below to the electricity utility price of Rwanda and the average of electricity utility price in Sub-Saharan Africa [55] [79] [62].

Peat_PP is a new technology in a mountainous country like Rwanda; therefore, it is not easy for anyone to fix the exact infrastructure requirement for the installation and commissioning. As a result, to weight the sensitivity of developing the Peat_PP technology, we increase the overnight cost from US\$2010/kW to US\$2500/kW and decrease it to US\$1500/kW. The same as the discount rate, the overnight increase causes the LCOE to increase. The plant factor changes from 91% to 80% also increases the LCOE and decreases to the electricity supply at secondary level to around 12% (see Fig. 12). It is preferable to improve the plant efficiency because the increase from 40% to 43% does not affect much the LCOE, however, it increases generation and reduce the CO₂ emissions [32].

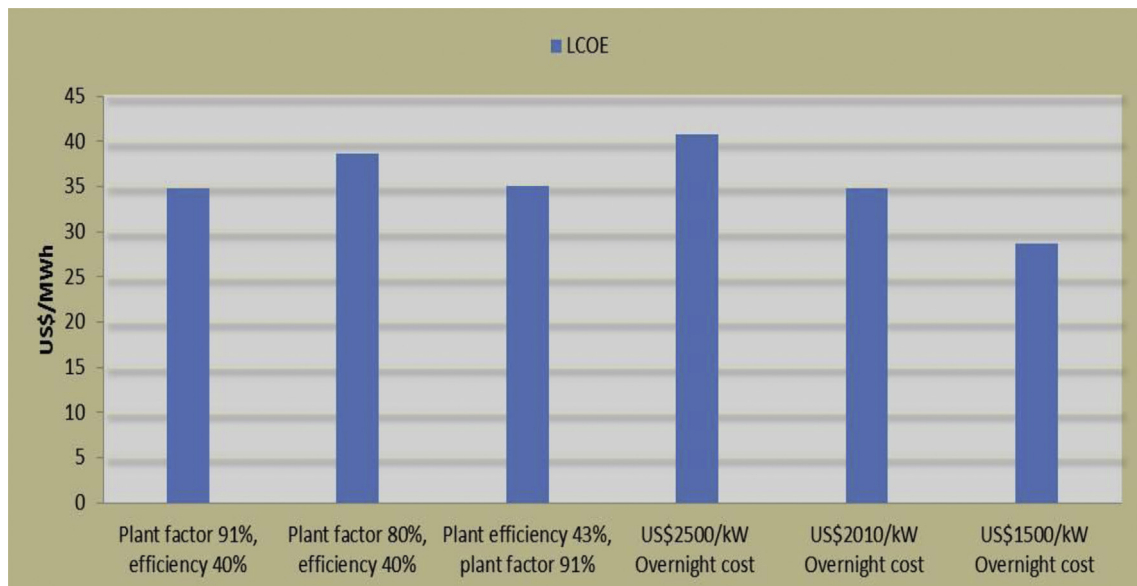


Fig. 12. Results of sensitivity analysis of LCOE of Peat_PP technology.

6. Conclusions and policy implications

We fully recognize that the availability of reliable electricity supply is not only a requirement for economic growth but also for social wealth and human development. Therefore, the access to the electricity services by all sectors of the economy is vital. The government of Rwanda has set two ambitious targets for electricity supply expansion to more than 1000 MW by 2017 and aggressive program to increase the access to the electricity for households from 4% in 2008 to 40% in 2025 [38] [77]. Conversely, the electricity supply expansion target was reduced to 563 MW by 2017–2018 after realizing the previous set target was not achievable [61], furthermore the electricity access target was also reviewed and set 48% by 2018 [39].

We totally support the Government of Rwanda energy policy and strategy, especially when it comes to its will to ensure the availability of sufficient, reliable and affordable energy supplies on defined target. However, after assessing three specific documents of energy policy and strategy elaborated from 2004 to 2013 [39] [41] [42], we notice that the emergency of energy infrastructure may be one of the reasons that was pushed the government to shorten the period of target without considering the real time of infrastructure development. This is supported by peat-to-power technology, which accounts 200 MW out of 563 MW targeted by 2017–2018, although, only 15 MW (Government investment) and 80 MW (private investment) are being developed [39], which means that it is impossible to meet the set target.

Accordingly, we estimated electricity demand input data basing to past demand statistics, economic development and population growth, and electricity supply referring to potential energy resources supply, overnight cost of energy infrastructure, real construction time of infrastructure, operation time, efficiency and plant life. In addition, we mention that the above explanation supports the reason why our results in 2017 does not correspond to 563 MW targeted by the government.

The Model for Analysis of Energy Demand (MAED) performs the modeling of the hourly load curve of Rwanda future electricity demand using and the date resulted serves as final demand input data into The Model for Energy Supply System Alternatives and their General Environmental Impacts (MESSAGE). MESSAGE optimizes Rwanda electricity generation according to the reference of energy supply scenario.

We observe that Rwandan electricity supply depends mainly on peat and natural gas, renewable energies such as local hydropower and regional hydropower, and geothermal. Peat power plant of 15 MW is being developed by the government and 80 MW by private investor are planned to be commissioned by 2017–2018 [20] [72], therefore, Peat-to-power is expected to boost the national energy system with a 26% and 40% share by 2017 and 2020, respectively. All along the study period, peat-to-power technology contributes more than 25% of total Rwanda electricity share. The total energy supply of electricity increases from 547.8 GWh in 2013 to 8488.6 GWh in 2045 at the secondary. The final electricity demand grows from 279 GWh in 2010 to 3361 GWh in 2045 for business as usual scenario electricity consumption forecast, and from 279 GWh in 2010 to 4940 in 2045, for high scenario of electricity consumption forecast. We therefore observe that during the study period the forecasted demand is secured at more than 30% of electricity supply margin.

For the perspective of installed capacity, from only 115 MW installed capacity in 2013 and increased to 215 MW in 2017, all energy supply technologies boost from 215 MW in 2017 to 521 MW installed capacity in 2030. From 2030, the electricity supply capacity is almost triple in 2045 and reaches 1705 MW. Peat_PP has a significant impact on the electricity supply between 2017 and 2045,

when 300 MW of power are fully installed at the end of the study period. We find that the introduction to the national grid of 30 MW of peat-to-power energy in 2017 helps phase out oil thermal power. As a result, an estimated cost of US\$65,000 per day required for the operation of oil thermal power plants can be stopped, the 25% of oil used in electricity generation is reduced to total oil imported and 40% of electricity price subsidies can also be removed [48] [54]. We also find that LCOE of all energy technologies conversion behavior remains under the range of the current electricity utility price except Solar_PP technology. The LCOE of Peat_PP technology is has less cost margin to the utility price of about 64%.

The MESSAGE model calculated the emissions from energy technologies, as based on emission factors calculated in previous section for Peat_PP and average emission factors of Gas_PP and Oil_PP technology. We observe that CO₂ emissions from Peat_PP increase from 96 ktonCO₂ in 2017 (94% of total CO₂ emissions) to 977.6 ktonCO₂ in 2045 (69% of total CO₂ emissions). Peat-to-power is identified to be a significant source of CO₂ emissions, the same as Gas_PP. It is important to note that the LCOE calculations include the carbon emission price. Although there is an increase in absolute emissions from 2025 to 2045, the study results reflect a possible reduction in carbon by around 30% if a new potential peat-to-power technology, such as upgraded hybrid peat types is used [80].

Acknowledgments

This work was supported by “Human Resources Program in Energy Technology” of the Korea Institute of Energy Technology Evaluation and Planning (KETEP), granted financial resource from Ministry of Trade, Industry and Energy, Republic of Korea. (Project No: 2015 4010 200820).

This work was also supported by the BK21+ program grant funded by the Korea government Ministry of Education.

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